

RAHUL AGARWAL

B.TECH CSE (IOT) STUDENT | FULL-STACK & AI/ML DEVELOPER

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Professional Summary

Full-stack and AI/ML developer pursuing a B.Tech in CSE (IoT), with experience architecting complete platforms spanning frontend (React.js/Vite), backend (Node.js, FastAPI), databases (MongoDB, SQLite, PostgreSQL), and embedded systems (ESP32). Built SmartVital, a multi-disease clinical risk prediction platform with explainable AI, and HackMatch, a real-time hackathon matchmaking platform with end-to-end encrypted chat. Comfortable owning a product from PRD through deployment.

Skills

- Languages:** C, Java, Python, JavaScript (ES6+)
- Web Development:** HTML5, CSS3, React.js, Vite, Responsive & Mobile-First Design, Frontend Development
- Frameworks / Backend / Databases:** Node.js, Express.js, FastAPI, Firebase, REST APIs, WebSockets, JWT Authentication
- Databases:** MongoDB, SQLite, PostgreSQL (Supabase)
- Tools & Platforms:** Git, GitHub, VS Code, Arduino IDE, Vercel, Netlify, Render, Canva
- Hardware & IoT:** ESP32, Arduino, IoT Systems & Sensor Integration, Real-Time WebSocket Communication
- AI & Machine Learning:** TensorFlow/Keras, SHAP, LIME, SMOTE, scikit-learn, Groq/LLM API, Prompt Engineering

Education

University of Engineering & Management (UEM), New Town	2025 – 2029
B.Tech in Computer Science & Engineering (IoT)	
Cumulative GPA: 8.07 / 10.00 (1st Semester)	
Sudhir Memorial Institute, Liluah	2025
Class XII (CBSE) Score: 68.4%	
M.C. Kejriwal Vidyapeeth	2023
Class X (ICSE) Score: 93%	

Projects

SmartVital – AI-Powered Multi-Disease Clinical Risk Prediction Platform	Dec 2025 – Present
Tech Stack: React.js, Vite, FastAPI, Python, MongoDB Atlas, ESP32, WebSocket, JWT, TensorFlow/Keras	
Built a full-stack clinical risk prediction platform using FastAPI, React/Vite, and MongoDB Atlas, with ESP32/WebSocket-based virtual IoT sensor integration for real-time vitals.	
Trained custom Artificial Neural Network models (Dense–Dropout–Sigmoid architecture) for four disease risk predictions, addressing class imbalance using SMOTE and weighted loss functions.	
Implemented SHAP and LIME explainability modules and a What-If simulator to make model predictions transparent and interpretable for clinical users.	
Designed role-based JWT authentication (patient/doctor/admin/researcher) and a conversational AI health assistant powered by Groq-hosted Llama 3.1.	
HackMatch – Real-Time Hackathon Team Matchmaking Platform	May 2026 – Present
Tech Stack: React.js, Node.js, Express.js, SQLite, JWT, WebSocket	
Built a real-time platform that matches hackathon participants with compatible teammates using a custom Synergy Score algorithm.	
Designed and shipped core product features (HackScore, Idea Board, Stack Clash) backed by a Node.js/Express REST API and SQLite database.	
Authored a full product requirements document (PRD) and deployment kit, and led diagnostic-first debugging of authentication and encryption issues.	
TRINETRA – AI-Powered Emergency Response System	December 2025
Tech Stack: ESP32, Python (Flask), Computer Vision, Web Dashboard	
Built an AI-driven safety ecosystem to reduce emergency response time within the critical 10-minute window.	
Integrated ESP32 wearable hardware, smartphone interface, and Flask-based AI backend.	
Implemented facial stress detection, real-time evidence capture, live GPS tracking, and automated SMS alert system.	
Optimized for low-cost deployment (< ₹500) for real-world scalability.	
AquaSense – Smart Irrigation & Environmental Monitoring System	March 2026
Tech Stack: ESP32, IoT Sensors, Web Dashboard, Machine Learning Models	
Developed an IoT-based precision agriculture system using ESP32 and multi-sensor integration.	
Automated irrigation decisions based on real-time soil moisture, temperature, humidity, and water flow data.	
Implemented AI modules for crop health classification, disease prediction, yield forecasting, and anomaly detection.	

Teaching Experience

Private Tutor (Part-Time), Self-Employed	2026 – Present
Teaching Class 10 students (15 students) in Computer Science with a focus on Java programming fundamentals, including data types, control structures, and problem-solving.	
Mentoring Class 8 students (5 students) in Physics, Chemistry, Biology, Computer simplifying concepts and improving conceptual understanding.	

Positions of Responsibility

Intern, IEDC Laboratory, UEM Kolkata	Summer 2026
Completed a summer internship in innovation and project development at UEM's IEDC Laboratory.	
Core Member – Web Development Team, Microsoft Student Society (MSS), UEMK	Dec 2025 – Present
Contributed to the development and maintenance of the society's official website, enhancing UI and overall user experience.	
Played a key role in organizing ChainSphere (Web3 Workshop), managing technical and coordination aspects.	
Awarded "Best Working Member" for outstanding contribution and active involvement.	
Web Development Member, Computer Society of India (CSI), UEMK	Jan 2026 – Present
Worked as part of the web development team, contributing to technical tasks and digital presence of the society.	
Assisted in organizing multiple technical and non-technical events, ensuring smooth execution and coordination.	

Achievements

Received Special Mention at Codexify for the project GoClever, recognizing innovation in building an AI-powered travel intelligence platform.	
Awarded "Best Working Member" at Microsoft Student Society (Web Development Team) for outstanding contribution, consistency, and impact in technical and organizational activities.	
Built RoutelQ (Next.js, Firebase, Leaflet Maps) at a hackathon hosted by Techno India Batanagar; also competed in Hackolution 2K26 and other collegiate hackathons.	